MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
ENGINEERING DESIGN & DRAFTING DIVISION

P&ID; DRAFTING GUIDE: INSTRUMENTATION & LINE SYMBOLOGY
DOC NO: **MRPL-DWG-STD-001** | REV: 05 | ANSI/ISA-5.1 STANDARD

1. INSTRUMENT LETTER CODE IDENTIFICATION (ISA-5.1)

Instrument tag names consist of two components: Functional Identification (letters) and Loop Identification (numbers). First letter identifies the measured process variable; subsequent letters denote the readout or output function:

1st Letter (Variable)	Meaning	Succeeding Letter	Meaning	Example Tag	Description
P	Pressure	T	Transmitter	PT-101	Pressure Transmitter (Electronic 4-20mA)
F	Flow Rate	CV	Control Valve	FCV-101	Pneumatic Flow Control Valve (Air to Open)
T	Temperature	I	Indicator (Local)	TI-1014	Bimetallic Dial Thermometer
L	Level	SLL	Switch Low Low	LSLL-102	Emergency Low Level Interlock Sensor
I	Interlock / Logic	—	Controller	I-101	DCS Logic Interlock Tripping Pump P-101

2. PIPING LINE SPECIFICATION CODING

All process piping tags follow standard format: **[Size]"-[Fluid Code]-[Line Number]-[Pipe Spec]-[Insulation]**

- **Size**: Nominal Pipe Size in inches (e.g. 4", 6", 2")
- **Fluid Service**: HC = Hydrocarbon, CW = Cooling Water, STM = Steam, N2 = Nitrogen, DR = Drain
- **Piping Class**: A1A = ASME Class 150 Carbon Steel, B1A = ASME Class 300, E1A = ASME Class 600
- **Insulation**: H = Hot Conservation (Mineral wool), C = Cold Insulation (Polyurethane), P = Personnel Protection